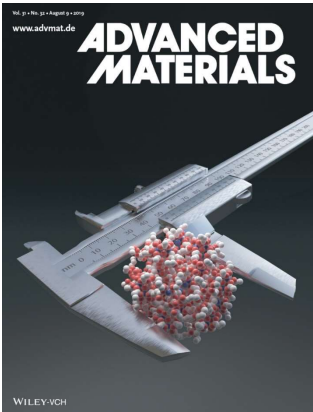


Unidad vi. Generación de NPs

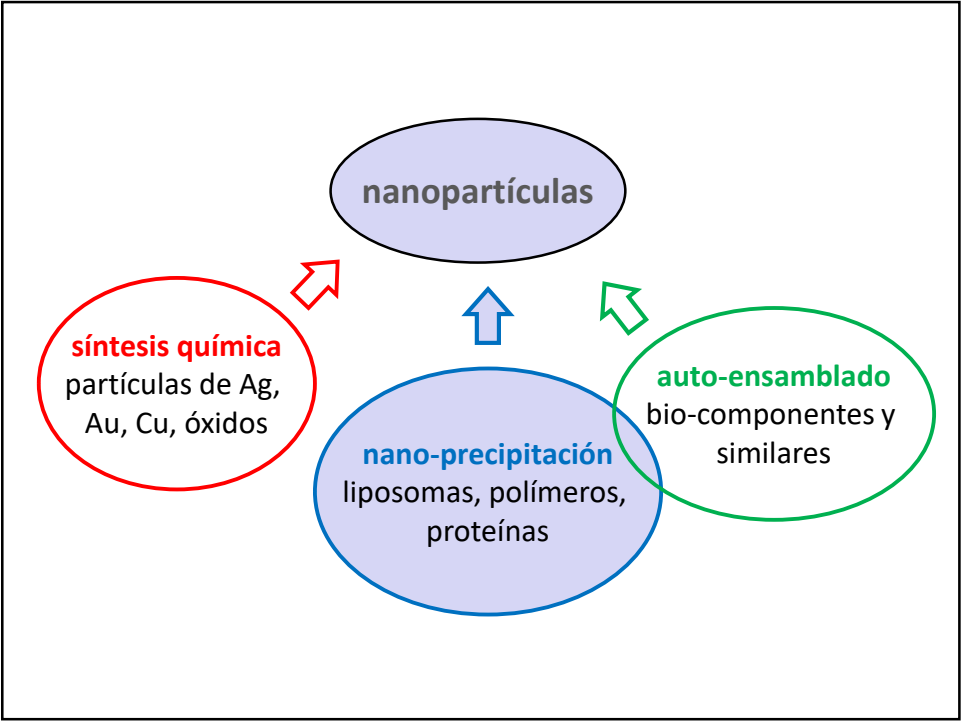


NANO partículas
1 - 500 nm

MICRO partículas
1 - 500 μm

Polymeric	Inorganic	Lipid-based
 Polymersome	 Silica NP	 Liposome
 Dendrimer	 Quantum dot	 Lipid NP
 Polymer micelle	 Iron oxide NP	 Emulsion
 Nanosphere	 Gold NP	
• Precise control of particle characteristics • Payload flexibility for hydrophilic and hydrophobic cargo • Easy surface modification • Possibility for aggregation and toxicity	• Unique electrical, magnetic and optical properties • Variability in size, structure and geometry • Well suited for theranostic applications • Toxicity and solubility limitations	• Formulation simplicity with a range of physicochemical properties • High bioavailability • Payload flexibility • Low encapsulation efficiency

Mitchell et al., *Nature Reviews: Drug Discovery* **20** (2021) 101





Ouzo: Grecia, Chipre
Arak: Siria, Líbano, Israel
Raki: Turquía, Balcanes
Anís: España
Pastis: Francia

CC(=C)C1=CC=C(C=C1)OC

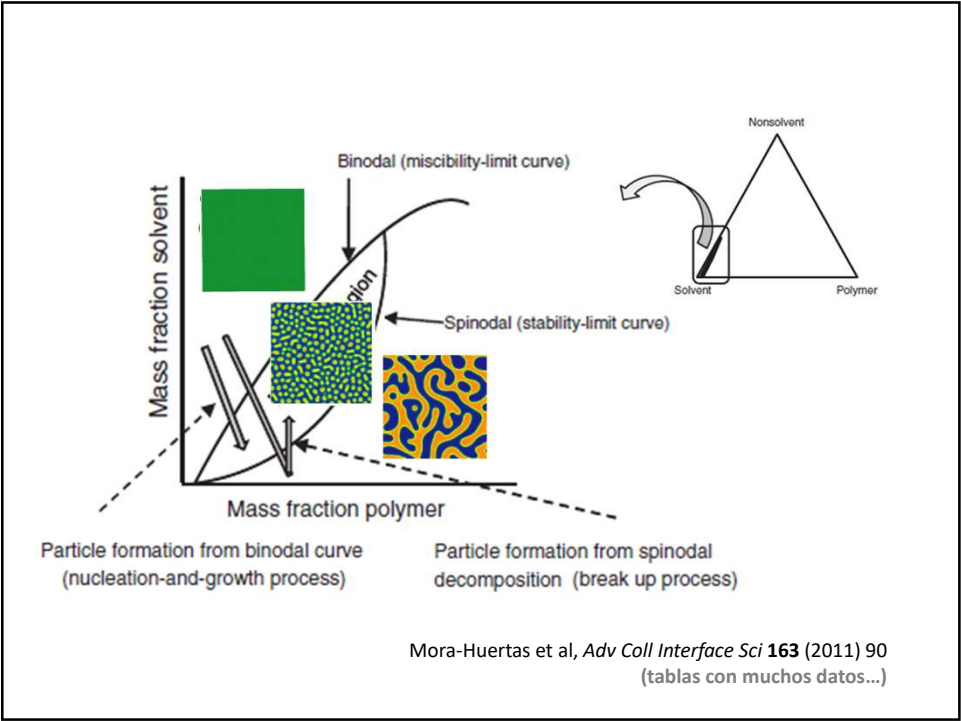
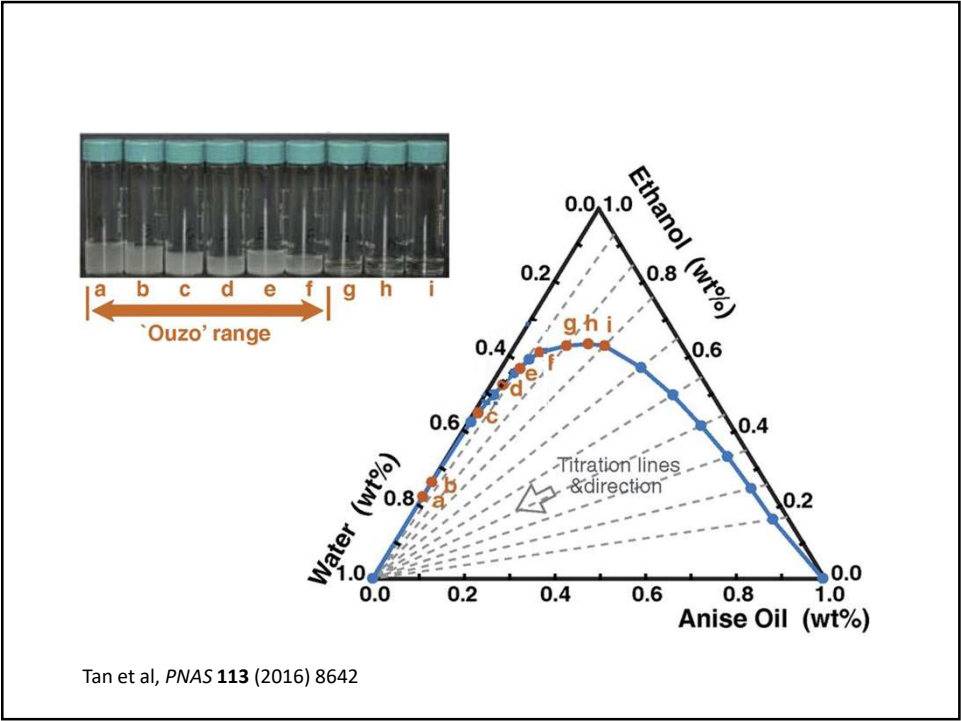
anetol

¿Qué se observa?

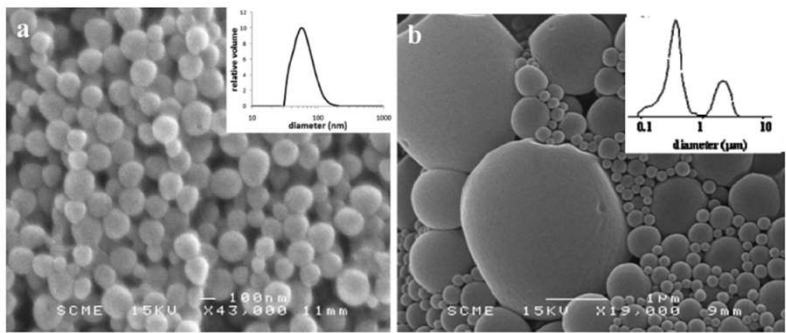
- Proceso espontáneo
- Dispersa la luz
- Estable a tiempos largos

Mezcla ternaria

Aceite de anís, 0.2%
Alcohol, 45%
Agua, 55%

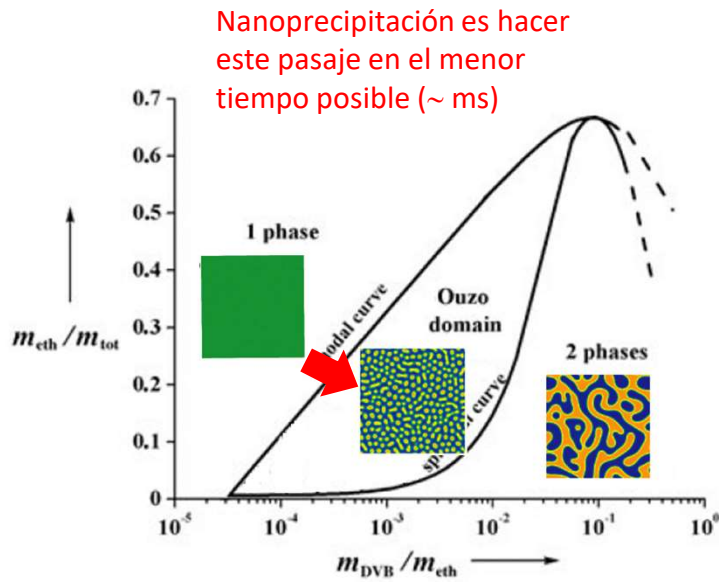


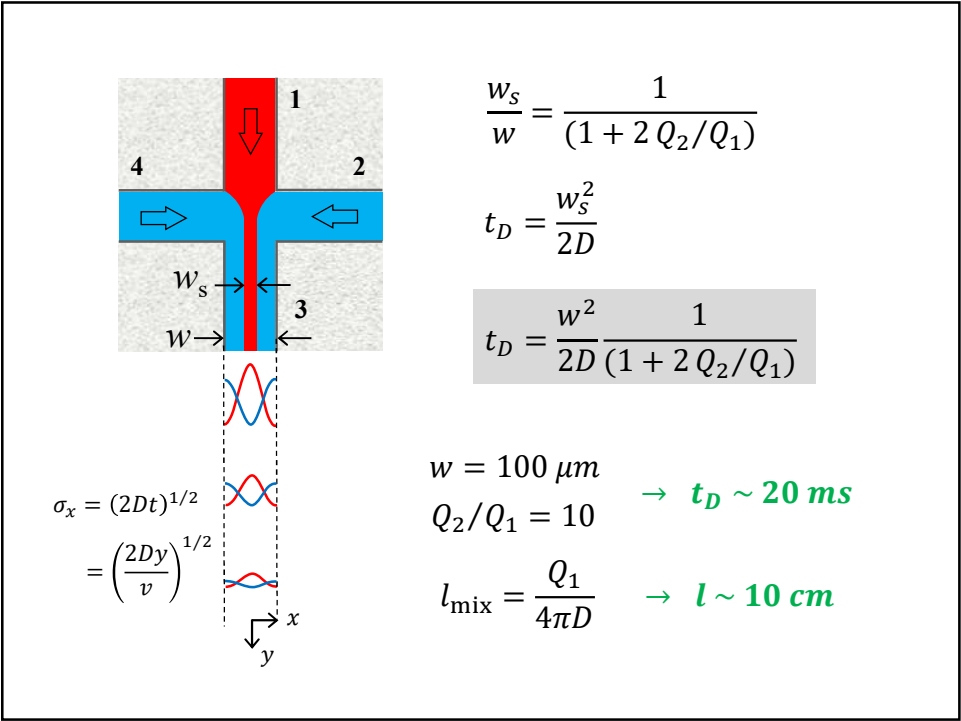
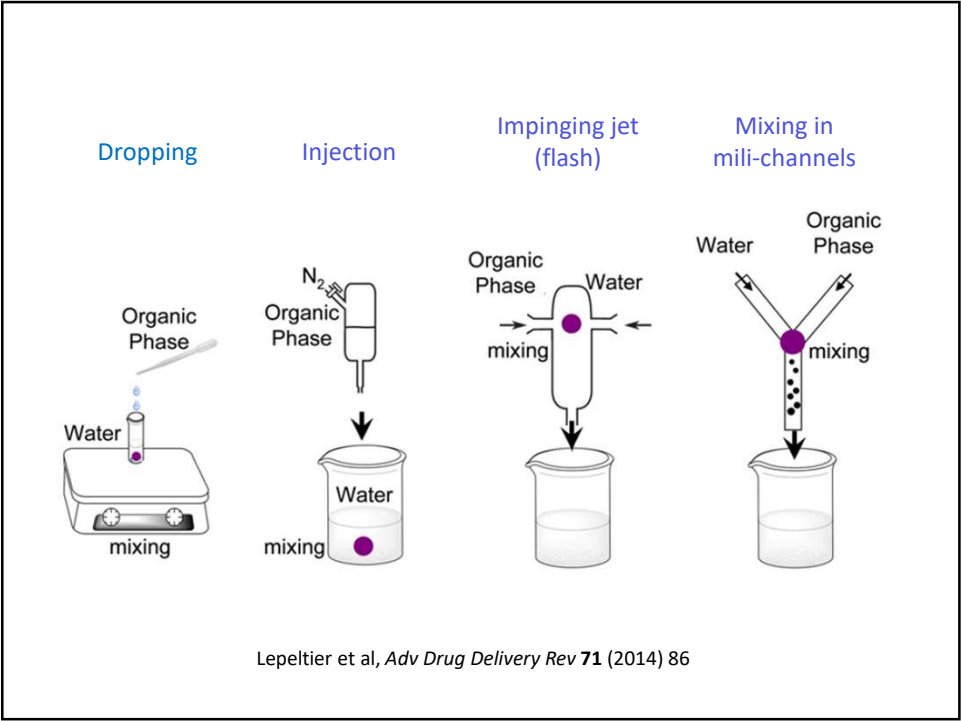
Objetivo: partículas monodispersas y tamaño controlado

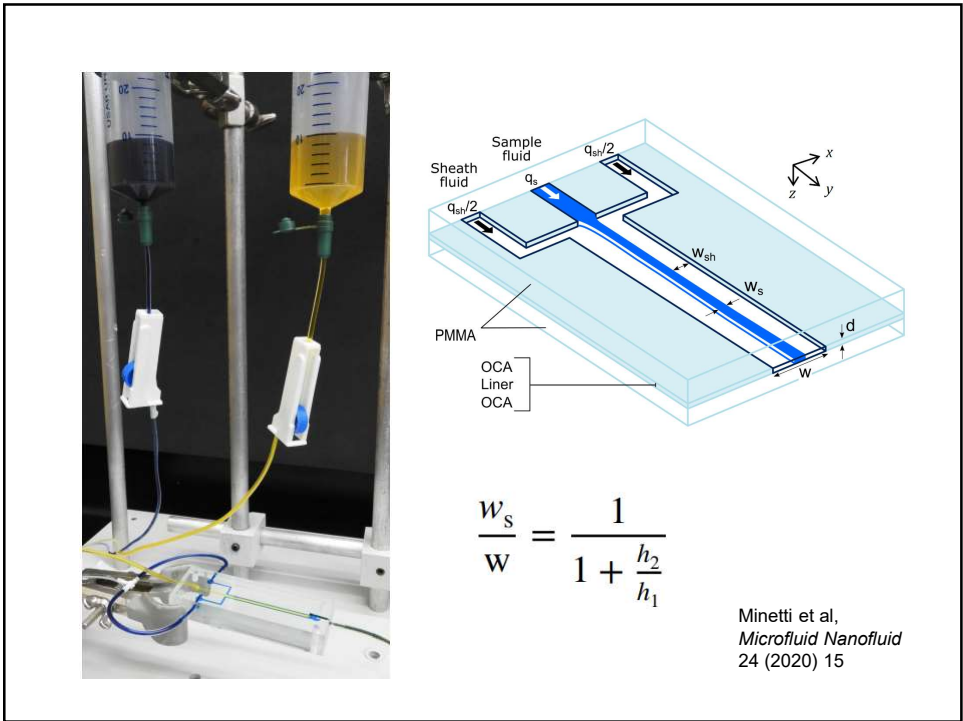
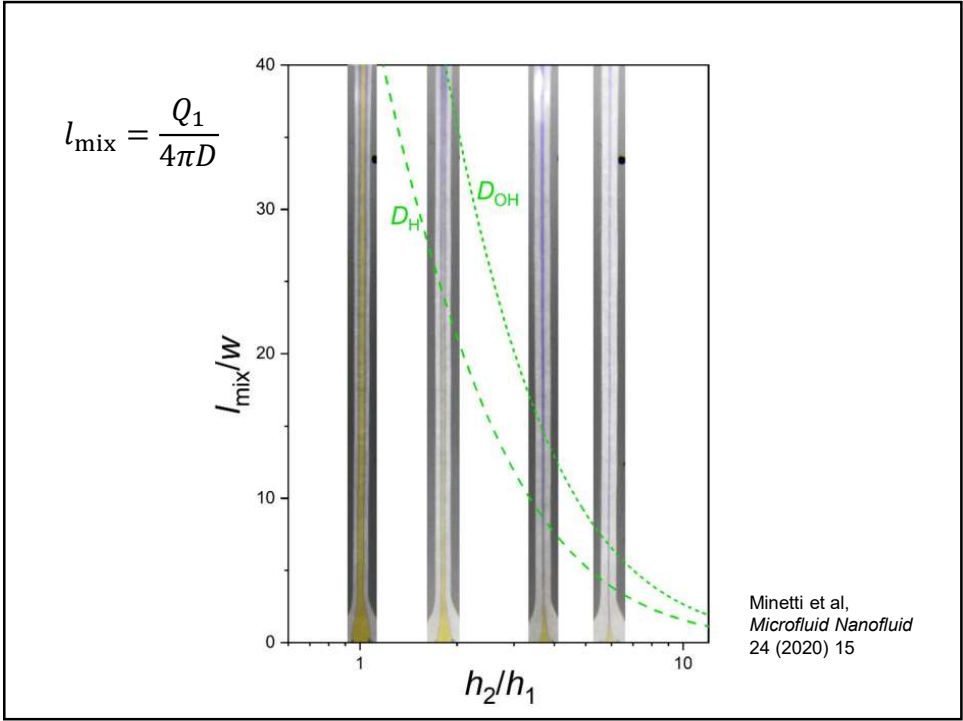


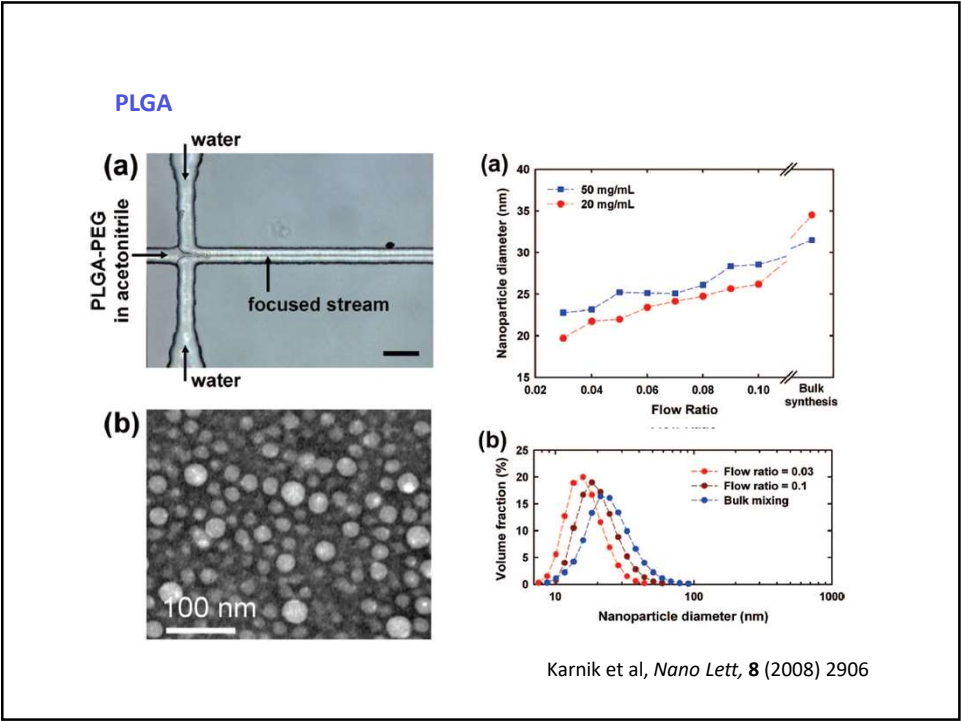
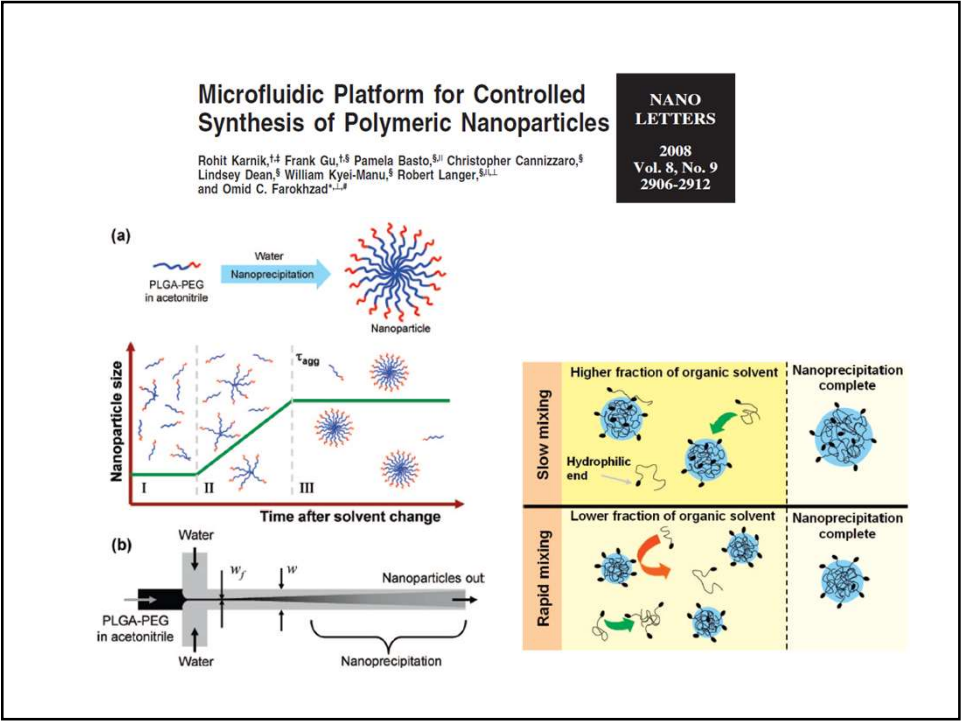
photographs of nanoparticles of PMMA synthesized in the Ouzo domain (a) and beyond the Ouzo boundary (b).

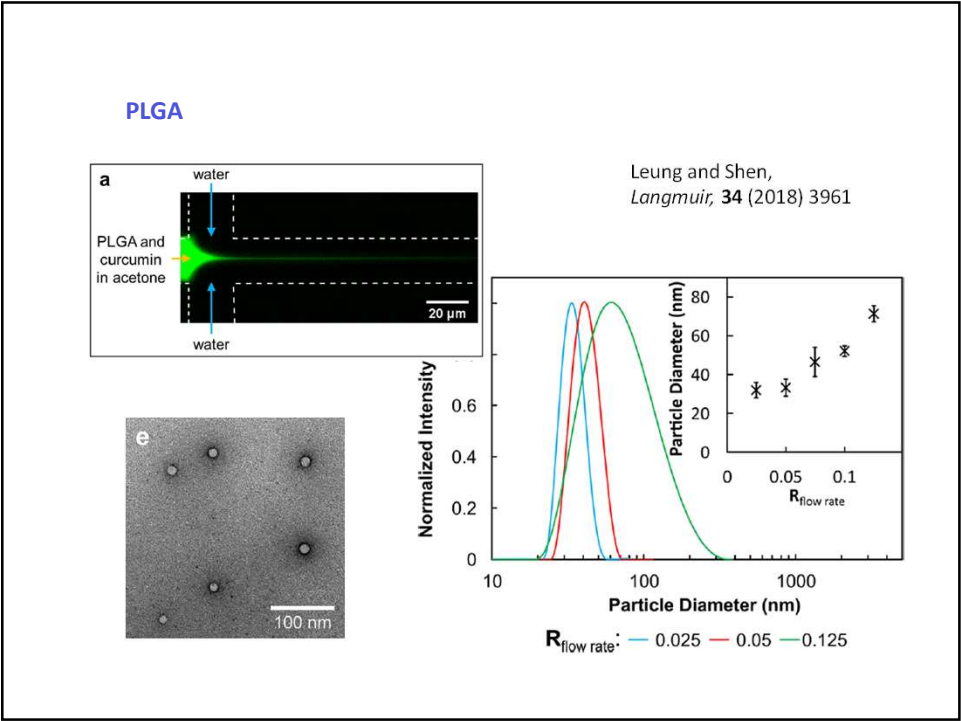
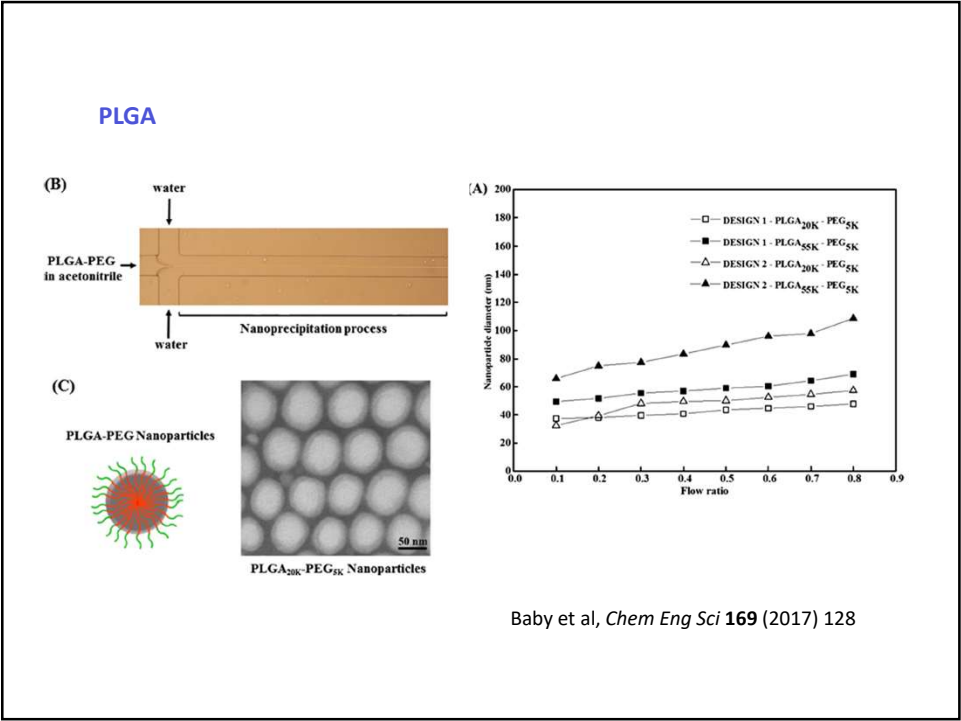
Aubry et al, *Langmuir* **25** (2009) 1970

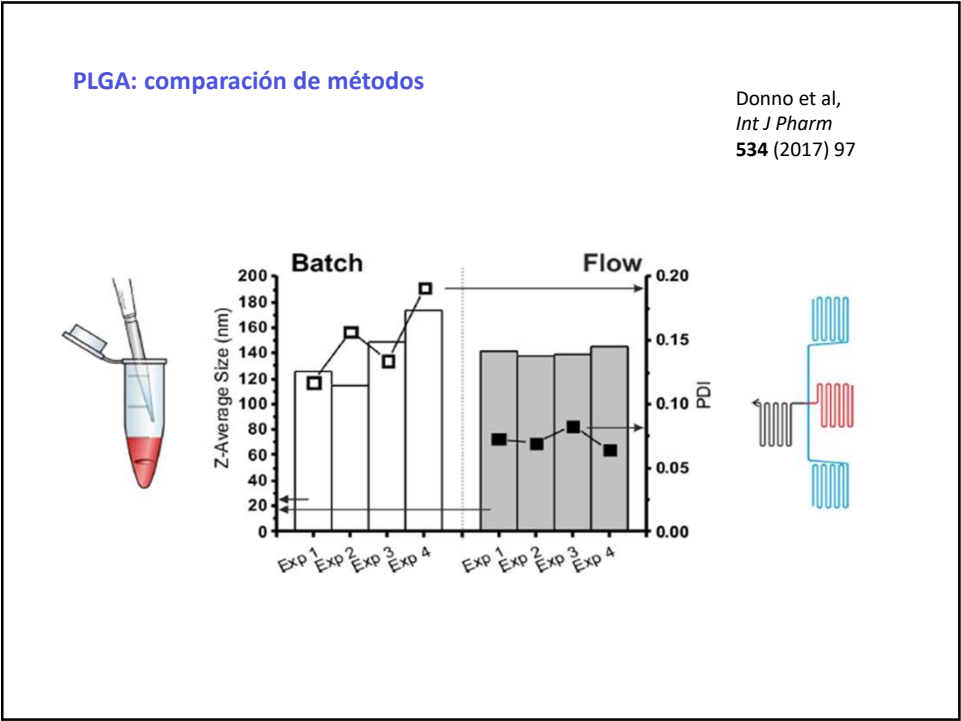
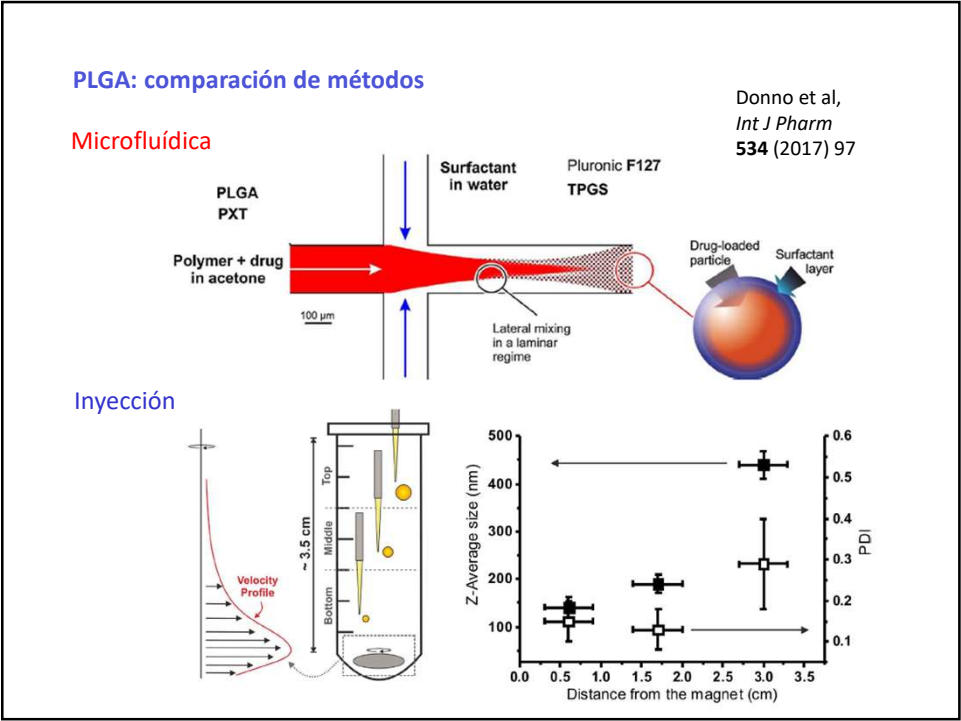


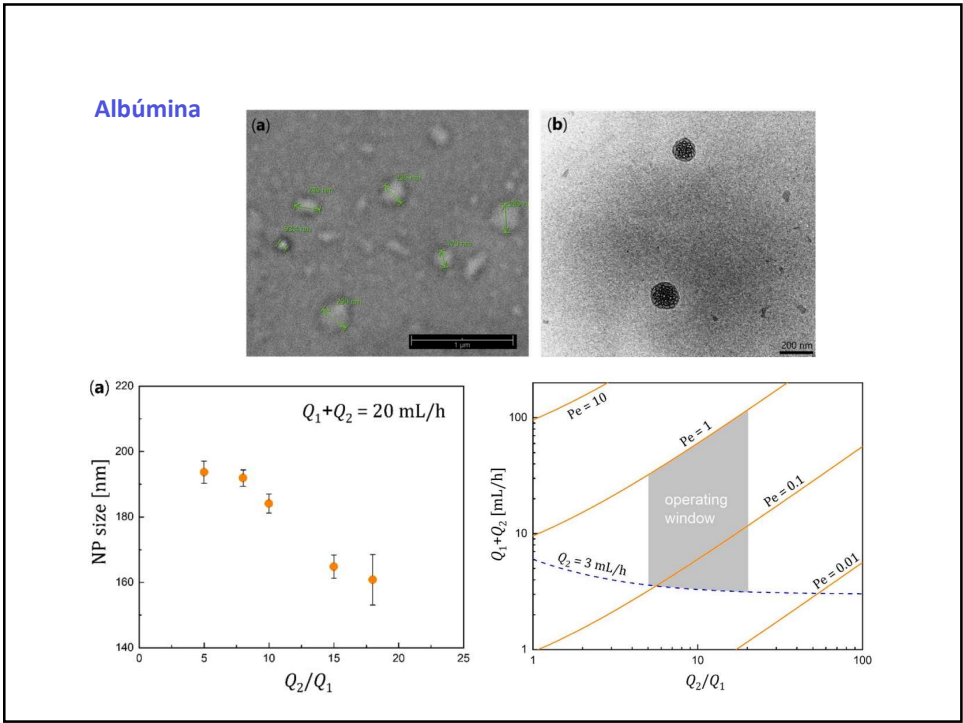
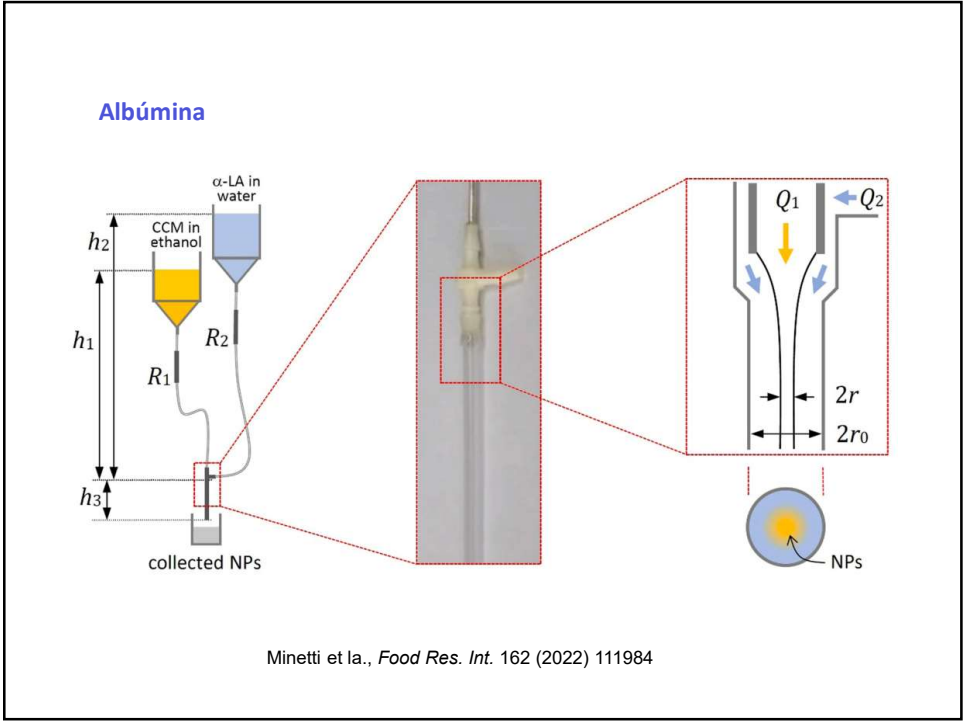




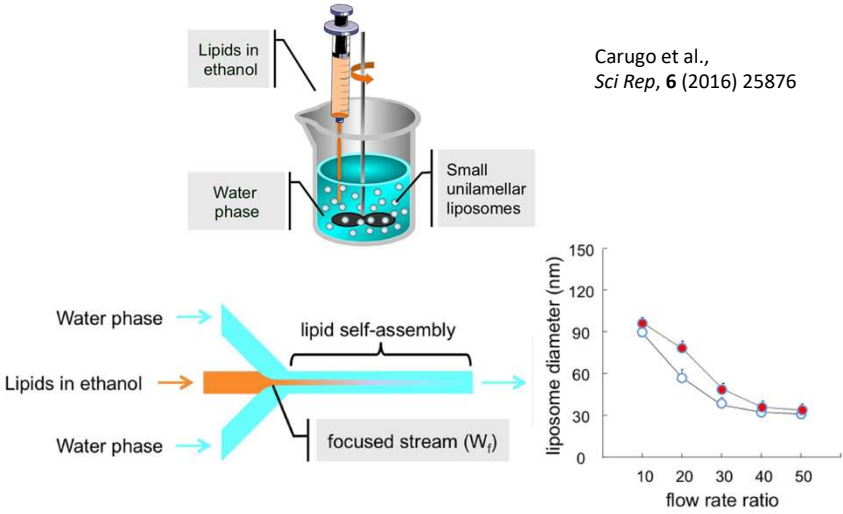








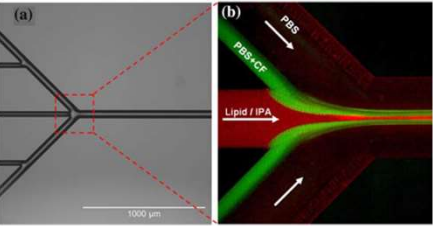
Liposomas



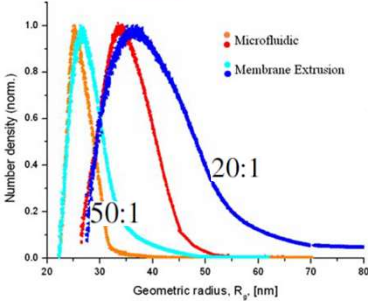
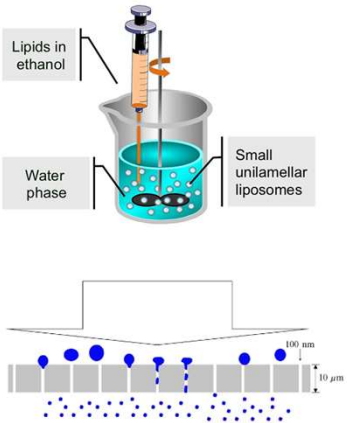
Carugo et al.,
Sci Rep, 6 (2016) 25876

Liposomas: comparación métodos

Microfluídica

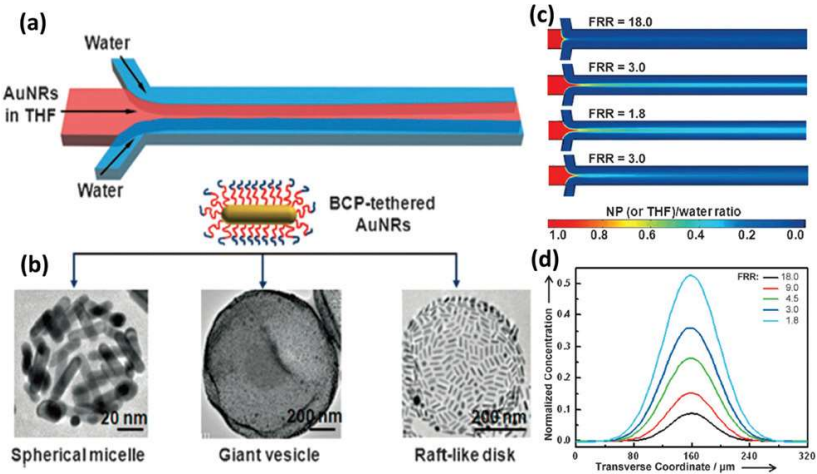


Inyección-extrusión



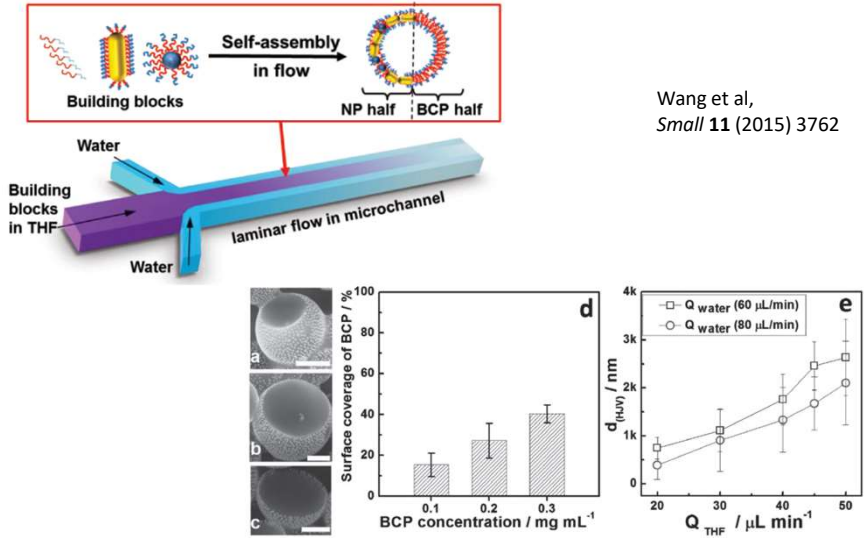
Jahn et al,
J Nanopart Res
10 (2008) 925

Auto-ensamblados



He et al., *Angew. Chem., Int. Ed.*, **52** (2013) 2463

Auto-ensamblados



Wang et al,
Small **11** (2015) 3762

